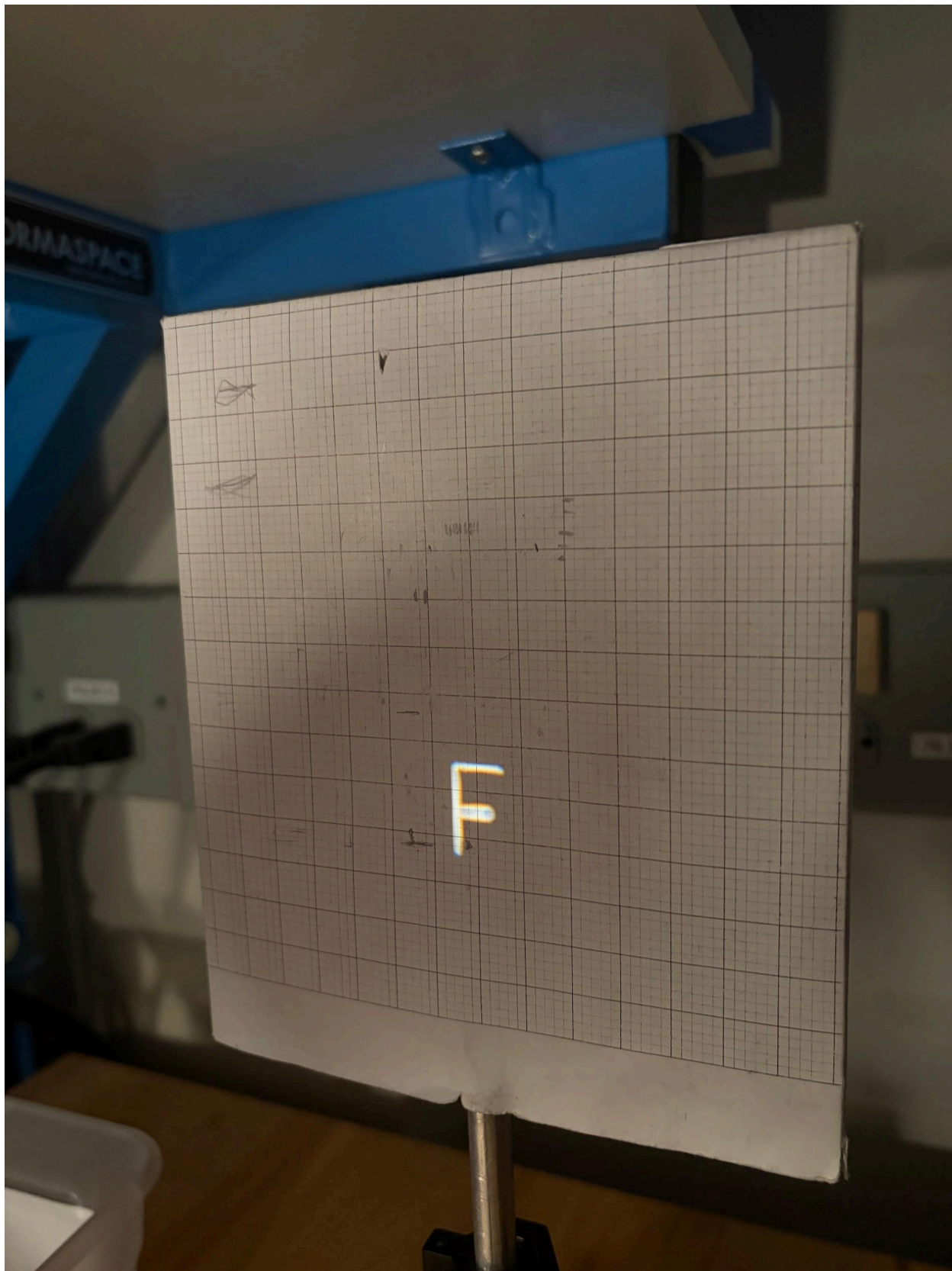
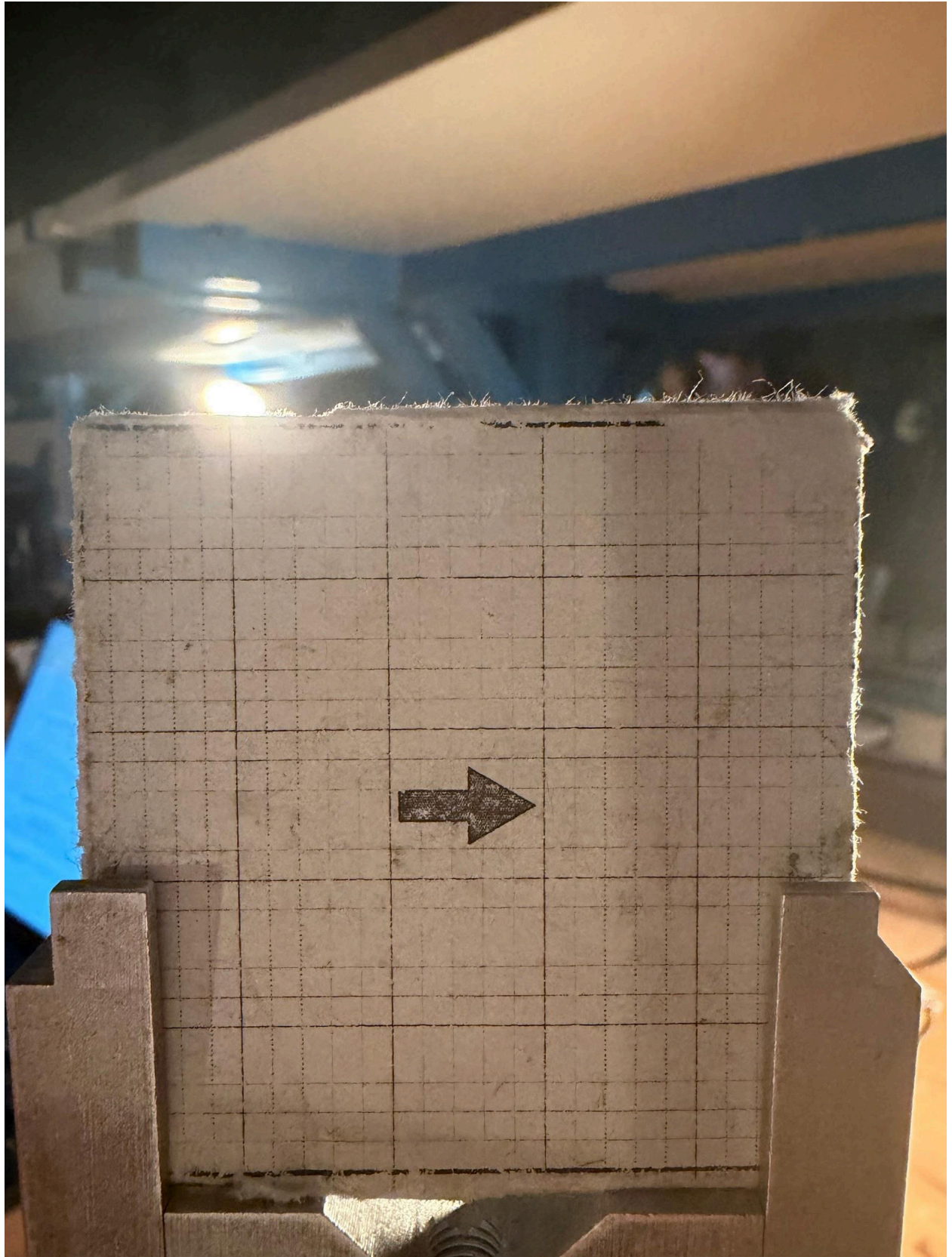
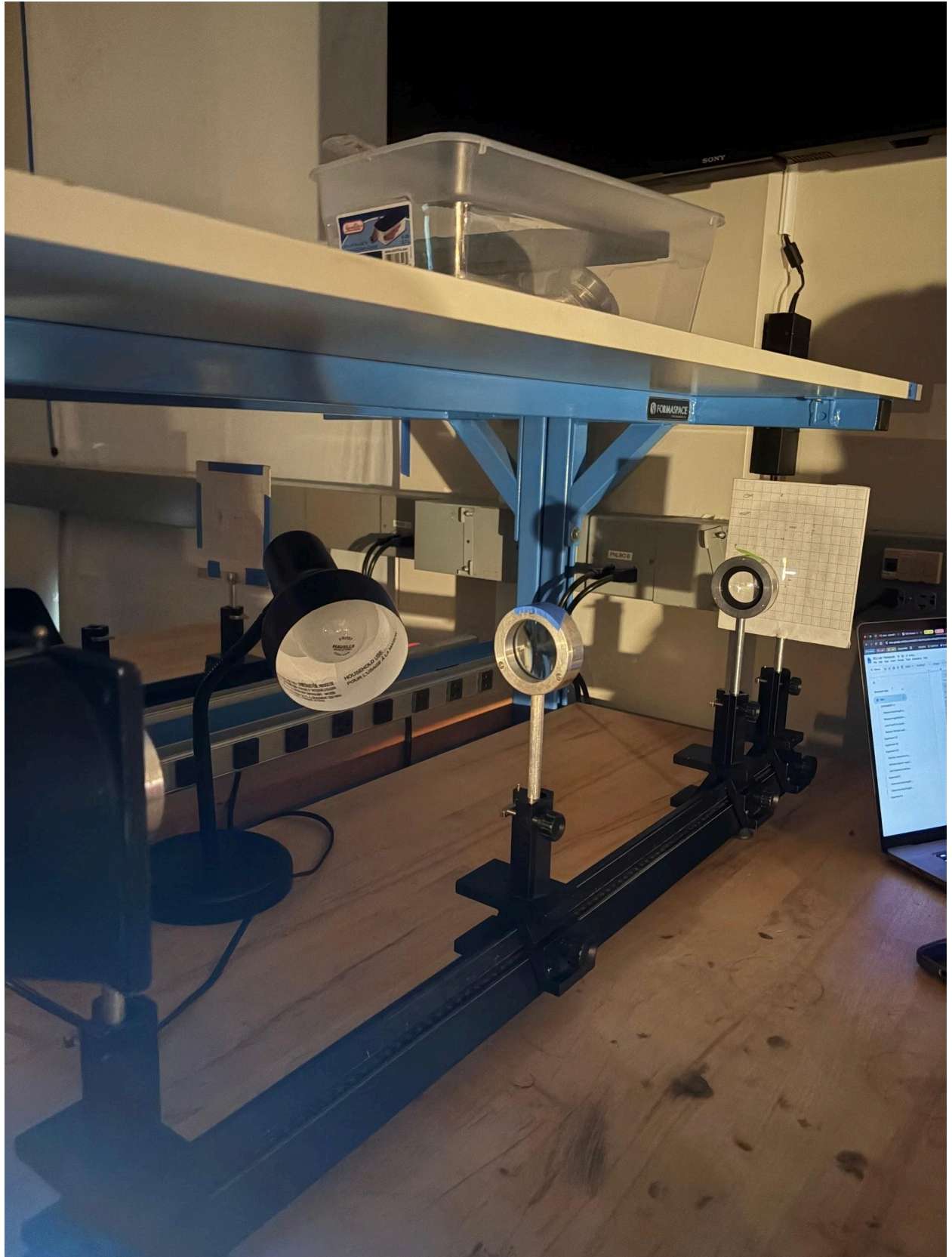












EXPERIMENT 1.1

Measure focal length of lenses [copy experimental procedure from Lab 0]

Gist is:

- Lens should be at infinity so put the lens as far away as possible

Lens A

Back wall position (cm)	Closest xi (cm)	Farthest xi (cm)
95.0	67.8	69.0
95.0	67.8	69.1
95.0	67.8	68.5

Lens B

Back wall position (cm)	Closest xi (cm)	Farthest xi (cm)
95.0	73.0	73.3
95.0	72.9	73.6
95.0	73.2	73.4

Lens C

Back wall position (cm)	Closest xi (cm)	Farthest xi (cm)
95.0	88.9	89.0
95.0	88.9	89.1
95.0	88.9	89.1

Measure magnification of lens C:

1. Measure position of F object
2. Measure height of F object hole
3. Move lens C and measure position
4. Measure closest back wall position to focus F upside down
5. Measure height of F on back wall at closest position
6. Measure farthest back wall position to focus F upside down
7. Measure height of F on back wall at farthest position
8. Repeat steps 3-7 for 3 total measurements

F object position: 17.4 ± 0.05 cm

F Object Height (cm):

2.230 ± 0.001	2.228 ± 0.001	2.224 ± 0.001
-------------------	-------------------	-------------------

Lens C

Lens C position (cm) ± 0.05	Closest Wall Position (cm) ± 0.05	Height F Closest (cm)	Farthest Wall Position (cm) ± 0.05	Height F Farthest (cm)
27.4	37.2	2.2 ± 0.1	38.6	2.8 ± 0.2
27.9	37.6	2.2 ± 0.1	37.9	2.4 ± 0.2
26.9	37.0	2.5 ± 0.1	38.2	2.7 ± 0.2

Lens B and C in series

13.2 $\pm$ 0.05 cm object

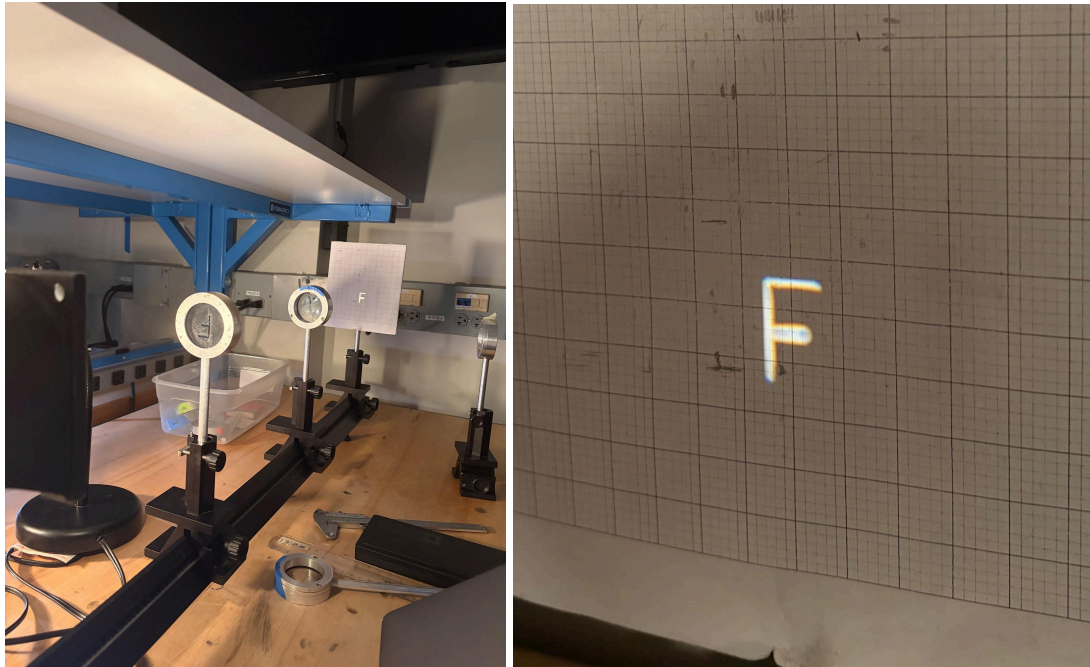
Lens C - 23.2 $\pm$ 0.05 cm

Screen would theoretically be at 33.2cm

Lens B Position (cm) $\pm$ 0.05	Closest Back wall (cm) $\pm$ 0.05	Farthest Back wall (cm) $\pm$ 0.05
63.2	94.6	96.2
62.7	95.9	97.0
63.7	95.7	96.4



Replace Old Back wall with F



Experiment 1.2

F object position: 13.55

Lens C position: 23.55

Lens B position: 28.55cm

Back wall position: 33.55cm

Therefore object distance = -5.0cm; image distance = 5.0cm. $M = 1.0$

Experiment 1.3

Measure F-object position, Lens B position, Lens A position, Screen position when the image is sharp, Original F-object height, Final image height on the screen

Object distance of Lens B = [Lens B] - [Object position]

Image distance of Lens A = [Back wall] - [Lens A]

[Object distance of A] = $(1/f_A - 1/di_A)^{-1}$

$di_B = (x_A - x_B) - do_A$

$M_{tot} = (di_B/do_B)(di_A/do_A)$

$M_{\text{tot empirical}} = \text{height on screen} / \text{og height}$

F object position (cm) pm 0.05	Lens B position (cm) pm 0.05	Lens A position (cm) pm 0.05	Backwall position (cm) pm 0.05	Final height on screen (cm) pm 0.1
13.5	24.1	49.8	75.2	3.1
13.5	26.9	55.5	75.5	2.7
13.5	27.9	55.3	74.1	2.6
13.5	23.3	48.0	74.1	3.2
13.5	21.0	40/2	76.2	4.7

Experiment 2.1

Find the nearpoint of eyeball

Bryan's right eye: 9.7, 10.0, 9.4 cm

Ines's right eye: 11.0, 10.6, 10.5 cm

Estimate angular magnification

Bryan: 1.5, Ines: 2.0

Use Camera to estimate angular magnification

Distance from camera to lens: 7.5 cm, 7.3, 7.2

Experiment 3

Determine focal length of double concave lens D

- Diverging lens alone cannot produce a real image
- If the diverging lens are placed close enough to the point where converging lens formed its image, then the diverging lens reduces the convergence but the rays still converge to form a real image further away, which we can project into a screen

Let

x_D : position of the concave lens.

x_0 : position of the image formed by the convex lens alone.

x_I : position of the final real image with both lenses present.

Theory:

Take the positive direction along the incoming light. The signed distances for the concave lens are

$$u = -(x_0 - x_D), \quad v = x_I - x_D.$$

Using the thin-lens equation,

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v},$$

or, writing $a = x_0 - x_D > 0$,

$$f = \frac{av}{a - v}.$$

A diverging lens should give $v > a$, so its focal length is negative.



Experimental procedure:

1. Adjust convex lens and screen until a sharp real image form. Fix the object and convex lens in place.
2. Measure x_0 - find uncertainty.
3. Insert concave lens before x_0 . Record its position as x_D .
4. Move the viewing screen to focus when both lenses present. Record its position as x_I .
5. Repeat the measurement multiple times, moving the screen + refocusing each time
6. Repeat for 6 different concave lens positions that give real images.

x F-object	x Lens B	x screen/initial image (best, worst)	x Lens D	X new image focusing position (best, worst)
13.5	60.0	83.2, 83.5	78.0	85.2, 85.8
13.5	60.0	83.2, 83.5	77.0	86.0, 87.5
13.5	60.0	83.2, 83.5	76.0	88.2, 88.7
13.5	60.0	83.2, 83.5	75.0	90.4, 91.9
13.5	60.0	83.2, 83.5	74.0	94.4, 94.9
13.5	60.0	83.2, 83.5	73.5	95.0, 97.0

X F-object, x convex, x concave all have uncertainties 0.005cm

X screen and x new image have uncertainties of best-worst / 2

Everything is measured in cm

We used lens B for convex lens

Determine focal length of spherical mirror E

A converging (concave) mirror forms a real image in front of the mirror, on the same side as the object, provided the object is beyond the focal point ($u > f$). The images can be focused onto a screen. Problem because screen blocks the light rays because on the same size, and the light blocks the screen if screen is placed behind. Our workaround is designing an experimental procedure where we can see the image focused on the original object (aided because it is opaque and black).

Experimental procedure:

1. Place F-object and measure its position.
2. Move mirror E until a sharp reflection is formed on the F-object.
3. Record closest and farthest position of mirror E.
4. Repeat a total of 6 trials.

F-object position (cm)	Mirror E Position Closest	Mirror E position Farthest
13.5	52.5	53.0
13.5	52.3	53.2
13.5	52.6	53.2
13.5	52.5	53.3
13.5	52.5	53.4
13.5	52.8	53.4

All measurements are in cm

F object position uncertainty is 0.005 cm

Mirror E position uncertainty is best-worst



Measuring Mirror E focal length: sharp image of F shown on F-object

Experiment 4

Determine angular magnification of compound microscope

X F object	X Lens B (objective)	X Lens C (eyepiece)	X Screen	Width on screen Pm 0.1
13.5	45.6	84.4	97.2	0.8
13.5	42.4	84.0	97.2	1.7
13.5	45.0	83.7	97.2	1.5
13.5	41.3	84.8	97.2	1.7

13.5	39.4	89.0	97.2	1.0
13.5	47.3	84.9	97.2	1.2

F width with vernier calipers: 0.944, 0.950, 0.940

Procedure:

1. Place F-object and measure its position.
2. Place lens B an arbitrary distance away, measure its location
3. Move the viewing plane until it focuses from lens B. Note this position
4. Place lens C behind the viewing plane roughly the length of its focal length. Record its position
5. Place the viewing plane such that an F focuses on the screen. Record its position
6. On the screen, record the width of the middle line of the F.
7. Repeat for a total of 6 times.

To calculate angular magnification via ratio of angular widths.

Angular width of object using small angle approx is $[F \text{ width}] / ([X \text{ screen}] - [X \text{ F object}])$

Angular width of image using small angle approx is $[\text{Width on screen}] / ([X \text{ Screen}] - [X \text{ Lens C}])$

We want to test relationship between s and angular magnification:

$s = [X \text{ lens C}] - [X \text{ lens B}] - [\text{focal length B}] - [\text{focal length C}]$

Should be linear relationship